Synoptic Question 27: The Physics of a Space Elevator

Reading Passage

Read the following article and then answer the questions that follow.

A space elevator is a proposed structure designed to transport payloads from a planet's surface into orbit without large rockets. The concept involves a incredibly strong cable extending from an anchor point on the Earth's equator to a massive counterweight located beyond geostationary orbit. Payloads would ascend the cable using mechanical climbers, powered either externally (e.g., by lasers) or internally.

The cable must be strong enough to support its own weight against Earth's gravity, the weight of the climbers, and the centrifugal force acting on its length due to Earth's rotation. To remain taut and stable, the center of mass of the entire structure (cable, counterweight, climbers) must be in geostationary orbit, meaning it orbits with the same period as the Earth's rotation. This requires the counterweight to be placed significantly beyond the geostationary altitude.

The primary challenge lies in the cable material. No conventional material is strong enough. Concepts often rely on materials like carbon nanotubes, which possess extremely high theoretical tensile strength. We can quantify material strength using these definitions:

- **Tensile Stress (σ):** The force applied per unit cross-sectional area of the material ($\sigma = F/A$). Its unit is Pascals (Pa).
- **Tensile Strain (ϵ):** The fractional change in length of the material under stress ($\epsilon = \Delta L/L_0$, where ΔL is the extension and L_0 is the original length). It is dimensionless.
- **Young's Modulus (E):** A measure of the material's stiffness, defined as the ratio of stress to strain within the elastic limit ($E = \sigma/\epsilon$). Its unit is Pascals (Pa).
- **Tensile Strength (σ_{max}):** The maximum stress a material can withstand before breaking or failing.

The cable experiences maximum stress near the geostationary altitude, where the combination of the weight below and the outward centrifugal force on the section above is greatest.

Data:

Data for Earth and a hypothetical space elevator design:

Table 1: Data for Space Elevator Analysis

Parameter	**Value**
Mass of Earth, M_E	5.97×10^{24} kg
Radius of Earth, R_E	6.37×10^{6} m
Gravitational constant, G	6.67×10^{-11} N m^2 kg^{-2}
Rotational period of Earth, T	24.0 hours
Mass of climber payload, m_c	20,000 kg
Average ascent speed of climber, v_{avg}	200 km h^{-1}
Cross-sectional area of cable, A	0.0015 m^2
Density of cable material, ρ_{cable}	1300 kg m^{-3}
Tensile strength of cable material, σ_{max}	6.0×10^{10} Pa
Young's Modulus of cable material, E	1.0×10^{12} Pa

Questions

(a) The space elevator cable extends to well beyond geostationary orbit.

  (i) Calculate the altitude h_{GEO} above the Earth's surface for a geostationary orbit.

  (ii) Calculate the orbital speed v_{GEO} of a satellite in geostationary orbit.

(b) Consider the forces acting on the cable itself. The cable extends from the surface ($r = R_E$) up to a maximum height r_{max}.

(i) Calculate the mass per unit length, μ, of the cable.

(ii) Explain qualitatively how the tension in the cable varies with altitude r from the Earth's center. Consider both the weight of the cable below r and the outward centrifugal force on the cable above r.

(iii) The tension is maximum near the geostationary altitude. A complex calculation shows this maximum tension F_{max} (supporting the lower part of the cable and the climber) is approximately 1.5×10^8 N when a 20-tonne climber is present. Calculate the tensile stress σ in the cable at this point.

(iv) Compare this stress to the tensile strength of the material and comment on the cable's safety margin.

(c) The climber ascends the cable.

(i) Calculate the average ascent speed in m s^{-1}.

(ii) Calculate the time taken, in days, for the climber to reach the geostationary altitude h_{GEO}.

(iii) Calculate the minimum work done against gravity to lift the 20,000 kg climber from the surface to h_{GEO}. (You will need to integrate or use the potential energy formula).

(iv) Calculate the average power required from the climber's propulsion system just to overcome gravity during the ascent (ignore kinetic energy changes and friction).

(d) The cable stretches under the immense tension.

(i) Using the maximum stress calculated in (b)(iii) and the Young's Modulus, calculate the tensile strain ϵ in the cable at the point of maximum tension.

(ii) If the entire cable length $L_0 \approx 100,000$ km experiences an average stress that is 50% of the maximum stress, estimate the total extension ΔL of the cable.

(e) Explain the purpose of the counterweight located beyond the geostationary altitude.

(f) Suggest one potential oscillation problem that a structure as long as a space elevator might face, and state the physics principle involved.

Solutions

(a) (i) For geostationary orbit, gravitational force = centripetal force. $T = 24 \times 3600 = 86400$ s. $\omega = 2\pi/T$.
$\frac{GM_E m}{r^2} = mr\omega^2 \implies r^3 = \frac{GM_E T^2}{4\pi^2}$
$r^3 = \frac{(6.67\times10^{-11})(5.97\times10^{24})(86400)^2}{4\pi^2} = 7.53 \times 10^{22}$
$r = 4.22 \times 10^7$ m (distance from Earth's center).
$h_{GEO} = r - R_E = (4.22 \times 10^7) - (6.37 \times 10^6) = 3.58 \times 10^7$ m.

(ii) $v_{GEO} = r\omega = (4.22 \times 10^7) \times \frac{2\pi}{86400} = 3070$ m s^{-1}.

(b) (i) Mass per length $\mu = \rho_{cable} \times A = (1300)(0.0015) = 1.95$ kg m^{-1}.

(ii) Below GEO altitude: Tension must support the weight of the cable below it (and any climber) plus provide the net force needed to keep that section rotating with the Earth. Weight component decreases with altitude ($g \propto 1/r^2$). Centrifugal force component ($mr\omega^2$) increases with altitude.
Above GEO altitude: The outward centrifugal force on a section of cable is greater than the inward gravitational force. This creates an outward tension supporting the sections below.
The tension increases from the anchor point up to a maximum near GEO altitude, then decreases further out towards the counterweight.

(iii) $\sigma = \frac{F_{max}}{A} = \frac{1.5\times10^8}{0.0015} = 1.0 \times 10^{11}$ Pa.

(iv) $\frac{\sigma}{\sigma_{max}} = \frac{1.0\times10^{11}}{6.0\times10^{10}} = 1.67$.
The calculated stress (100 GPa) exceeds the tensile strength (60 GPa). This cable design would fail. (Note: A tapered cable, thicker at GEO, is needed). If we assume the tensile strength was $\sigma_{max} = 1.6 \times 10^{11}$ Pa, the safety margin would be $1.6/1.0 = 1.6$.

(c) (i) $v_{avg} = 200$ km h$^{-1} \times \frac{1000}{3600} = 55.56$ m s^{-1}.

(ii) Time $t = \frac{\text{distance}}{v_{avg}} = \frac{h_{GEO}}{v_{avg}} = \frac{3.58\times10^7}{55.56} = 644300$ s.
$t(\text{days}) = \frac{644300}{24\times3600} = 7.46$ days.

(iii) Work done = Change in GPE = $GPE_{final} - GPE_{initial}$
$W = \left(-\frac{GM_E m_c}{r_{final}}\right) - \left(-\frac{GM_E m_c}{r_{initial}}\right)$
$r_{initial} = R_E = 6.37 \times 10^6$ m. $r_{final} = r_{GEO} = 4.22 \times 10^7$ m.
$W = GM_E m_c \left(\frac{1}{R_E} - \frac{1}{r_{GEO}}\right)$
$W = (6.67 \times 10^{-11})(5.97 \times 10^{24})(20000) \left(\frac{1}{6.37\times10^6} - \frac{1}{4.22\times10^7}\right)$
$W = (7.96 \times 10^{18}) \times (1.570 \times 10^{-7} - 0.237 \times 10^{-7}) = (7.96 \times 10^{18}) \times (1.333 \times 10^{-7})$
$W = 1.06 \times 10^{12}$ J.

(iv) Average Power $P_{avg} = \frac{W}{t} = \frac{1.06\times10^{12}}{644300} = 1.65 \times 10^6$ W (or 1.65 MW).

(d) (i) $E = \sigma/\epsilon \implies \epsilon = \frac{\sigma}{E}$
$\epsilon = \frac{1.0\times10^{11}}{1.0\times10^{12}} = 0.10$.

(ii) Average stress $\sigma_{avg} = 0.50 \times \sigma_{max} = 0.50 \times (1.0 \times 10^{11}) = 5.0 \times 10^{10}$ Pa.
Average strain $\epsilon_{avg} = \frac{\sigma_{avg}}{E} = \frac{5.0\times10^{10}}{1.0\times10^{12}} = 0.05$.
$L_0 = 100,000$ km $= 1.0 \times 10^8$ m.
$\Delta L = \epsilon_{avg} \times L_0 = 0.05 \times (1.0 \times 10^8) = 5.0 \times 10^6$ m (or 5000 km).

(e) The counterweight is placed beyond geostationary orbit so that the outward centrifugal force ($F_c = mr\omega^2$) acting on it is significantly greater than the inward gravitational force ($F_g = GM_E m/r^2$). This net outward force provides the tension needed to keep the entire cable taut and support the weight of the cable sections below geostationary orbit. It ensures the center of mass of the whole system remains at geostationary altitude.

(f) Potential oscillation: Transverse oscillations (like a guitar string). Principle: Resonance. If the cable is subject to a periodic driving force (e.g., wind, Coriolis forces due to climbers moving, gravitational influence of the Moon) whose frequency matches one of the cable's natural frequencies of vibration, the amplitude of these transverse oscillations could build up dramatically, potentially leading to instability or failure.